

Environmental and Coastal Correlates of Whale Shark Movement Persistence

A Spatial GAMM Analysis of Irregular Argos Satellite Tracking Data

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Abstract

Whale shark movement is shaped by heterogeneous environmental conditions, but the factors associated with movement persistence remain incompletely understood. We examined the association between environmental and coastal covariates and whale shark movement persistence using irregular satellite tracking data. Individual tracks were first regularized by fitting a state space model with a 24 hour time step to obtain daily predicted locations, from which movement persistence was estimated. Daily locations were then linked to sea surface temperature, bathymetry, distance to the nearest coast, and coast type. We modeled movement persistence using a spatial generalized additive mixed model with standardized continuous covariates, a two dimensional smooth of latitude and longitude, and shark level random effects. Movement persistence was positively associated with sea surface temperature, shallower bathymetry, and greater distance to the nearest coast. Coast type was not significant as a main effect, but the association between distance to coast and movement persistence differed significantly between island and continental coasts. Strong residual spatial structure remained after adjustment for measured covariates. These results suggest that whale shark movement persistence varies with both local environmental conditions and broader spatial context, and they support the use of spatial mixed modeling frameworks for linking marine telemetry data to environmental drivers of movement behaviour.

1 Introduction

Whale sharks (*Rhincodon typus*) are highly mobile marine animals that move across a wide range of ocean environments, and their space use is likely shaped by environmental and coastal conditions. Understanding the drivers of whale shark movement is therefore important for characterizing habitat use and broader patterns of spatial behaviour. Satellite telemetry has greatly expanded the study of marine animal movement, but inference from raw tracking data is often complicated by substantial location uncertainty and irregular sampling through time. In Argos telemetry data, location error can be considerable, and analyses that ignore this uncertainty may be inappropriate for movement studies (1). More generally, ecological tracking time series often require model-based processing before behavioural quantities can be estimated consistently and interpreted reliably (2; 3). State-space models are well suited to this setting because they separate the latent movement process from observation error and provide a principled framework for reconstructing animal trajectories from imperfect telemetry records (2; 3).

A useful summary of movement behaviour derived from such trajectories is movement persistence, which quantifies the extent to which movement remains directed over time. Higher persistence corresponds to more sustained directional travel, whereas lower persistence reflects less directed movement. Because movement persistence varies continuously along a track, it provides a convenient way to study how behaviour changes across environmental gradients. Previous work has shown that movement persistence can be modeled as a function of environmental predictors and can reveal important movement-environment relationships in marine species (4). However, although environmental influences on marine animal movement have been widely studied, the environmental correlates of whale shark movement persistence remain less well understood.

In this study, irregular Argos satellite tracks of whale sharks in the eastern Pacific were first regularized to daily predicted locations using a state-space model, after which movement persistence was estimated from the reconstructed trajectories. Daily locations were then linked to sea surface temperature, bathymetry, distance to the nearest coast, and coast type. A spatial generalized additive mixed model was used to evaluate associations between these covariates and movement persistence while accounting for residual spatial structure and repeated observations within sharks. This framework allowed environmental and coastal correlates of whale shark movement persistence to be assessed on a common daily scale using telemetry data that were noisy, irregular, and spatially structured.

2 Methods

2.1 Satellite tracking data

The raw dataset consisted of spatio-temporal Argos satellite telemetry observations from 60 tagged whale sharks collected between 2011 and 2021. Each observation included an individual identifier, timestamp, latitude, longitude, Argos location class, and the satellite receiving the transmitted signal. The tracking records covered the eastern Pacific region, with observations concentrated off the Pacific coasts of Central and South America and extending westward into the open ocean. The telemetry data were inherently irregular in time, with observation intervals varying substantially both within and across individuals. Figure 1 shows the spatial distribution of the retained telemetry locations across the eastern tropical Pacific study region.

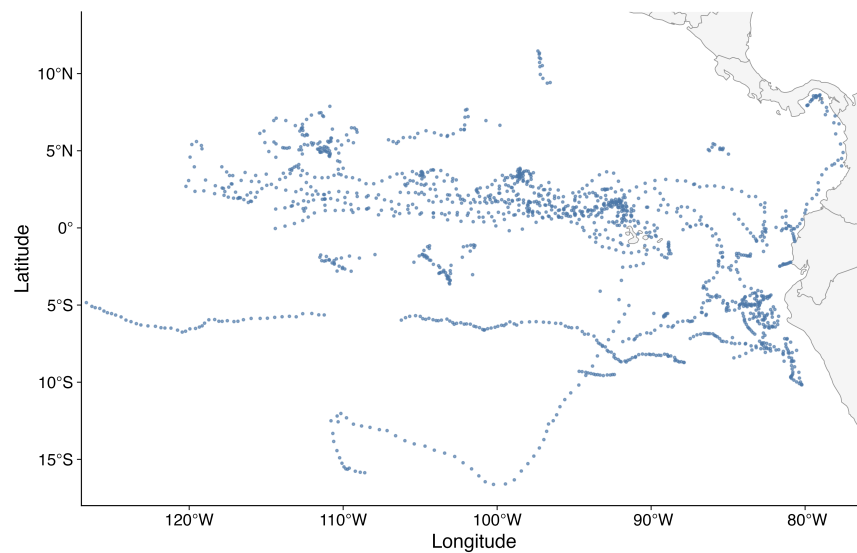


Figure 1: Spatial distribution of retained whale shark telemetry locations across the eastern tropical Pacific study region after initial filtering. Points show locations that remained after application of the movement-processing threshold, prior to track regularization. The retained records were concentrated along the Pacific coasts of Central and South America, with additional observations extending into offshore waters. This figure provides an overview of the spatial extent and geographic coverage of the telemetry data used in the subsequent track-processing steps.

2.2 Data cleaning and track segmentation

Prior to movement modeling, records with invalid temporal information were removed. Specifically, 15 observations with missing timestamps were excluded, while no missing or invalid longitude or latitude values were detected. The data were then checked for duplicate records and temporal consistency within individuals. One duplicated timestamp was identified for shark 203645 and removed, and all trajectories were ordered to ensure strictly increasing timestamps. After cleaning, the dataset contained 12,157 observations from 60 individual sharks.

Because the telemetry data were highly irregular in time, individual trajectories were divided into separate segments whenever the gap between two consecutive observations exceeded seven days. Only segments with at least 20 observations were retained for subsequent modeling. This segmentation strategy was used to avoid treating extended periods without detections as continuous movement and to retain track sections with sufficient temporal continuity for state-space estimation. A unique segment identifier was created by combining shark ID and segment number.

2.3 State-space modeling and daily predicted locations

After segmentation and filtering, 58 segments from 32 individual whale sharks were retained for state-space modeling. Track segments were regularized using a continuous-time correlated random walk state-space model implemented with the `fit_ssm()` function in the `aniMotum` package in R. This framework accounts for irregular sampling intervals while incorporating Argos location uncertainty through the observation process. In this formulation, the observed location $\mathbf{y}_t$ is treated as an error-prone measurement of an underlying latent location $\mathbf{x}_t$, such that

$$\mathbf{y}_t = \mathbf{x}_t + \boldsymbol{\varepsilon}_t,$$

where

$$\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_t),$$

and Σ_t denotes the observation error covariance matrix at time t .

Models were fitted using a 24-hour time step and a biologically motivated maximum speed threshold of 2.7 m/s. The daily time step was chosen to reflect movement at a biologically meaningful scale while also matching the temporal resolution used for subsequent extraction of environmental covariates. Model output consisted of one predicted latitude, longitude, and timestamp per day for each retained segment. These regularized daily predicted locations were then used for estimation of movement persistence and linkage with environmental and coastal covariates. The state-space formulation treats observed Argos locations as error-prone realizations of an unobserved movement path and assumes that the latent trajectory can be represented by a continuous-time correlated random walk. Such a formulation is appropriate for telemetry records with irregular sampling intervals, where direct use of the raw observations would not support consistent day-to-day linkage with environmental covariates. Regularization to a daily time step provides a common temporal scale for subsequent analysis, although prediction at that scale may smooth over short-term movement variation present in the original tracks.

2.4 Estimation of movement persistence

Movement persistence was estimated from the regularized daily predicted trajectories using a movement persistence model implemented with the `fit_mpm()` function in the `aniMotum` package in R. In this framework, the velocity vector at time t depends on the velocity vector at the previous time step according to

$$\mathbf{v}_t = \gamma_t \mathbf{v}_{t-1} + \boldsymbol{\eta}_t, \quad \gamma_t \in (0, 1),$$

where $\mathbf{v}_t$ denotes the velocity vector at time t , γ_t is the movement persistence parameter, and $\boldsymbol{\eta}_t$ is a random error term. Larger values of γ_t indicate more persistent and directed movement, whereas smaller values indicate less directed movement. Estimated movement persistence values were extracted from the fitted model output for converged trajectories and used as the response variable in subsequent environmental analyses. Estimation of movement persistence further assumes that behaviour can be summarized through temporal dependence in successive velocity vectors and that this dependence is meaningfully defined on a regular daily scale. Use of daily predicted locations therefore provides the temporal consistency required for comparable persistence estimates across individuals. At the same time, movement persistence should be interpreted as a continuous measure of directional consistency, not as a full classification of behavioural state.

2.5 Environmental covariates

Environmental covariates were linked to each regularized daily predicted location derived from the state-space model. Sea surface temperature (SST) was extracted from the NOAA Optimum Interpolation Sea Surface Temperature daily product (`ncdcOisst21Agg_LonPM180`) using the `rerddap` and

`rerddapXtracto` packages in R. For each predicted shark location, SST was matched by longitude, latitude, and date so that each daily movement estimate was paired with an SST value at the corresponding spatial and temporal position. Timestamps were converted to calendar dates prior to extraction, and SST values were retrieved iteratively by year across the study period.

Bathymetry was extracted separately for each regularized daily predicted location using the `etopo180` dataset, also accessed through `rerddap` and `rerddapXtracto`. Bathymetric values were obtained with the `rextracto()` function using the `altitude` parameter and a spatial extraction window of 0.1° in both longitude and latitude. The resulting bathymetry values were merged into the analysis dataset at the observation level.

Observations with positive bathymetry values were excluded prior to analysis because they indicated locations above sea level and were therefore biologically implausible for whale sharks. To characterize coastal context, two additional spatial covariates were derived for each daily predicted location: distance to the nearest coast and coast type. Coastline geometry was obtained from Natural Earth land polygons. For each predicted shark location, the nearest land polygon was identified using `st_nearest_feature()`, and distance to coast was computed using `st_distance()` and converted to kilometers. Coast type was defined by classifying the nearest coastline as either island or continental. Polygons intersecting a predefined island bounding box ($x_{\min} = -92.5$, $x_{\max} = -89.3$, $y_{\min} = -1.8$, $y_{\max} = 0.8$) were classified as island coasts, whereas all remaining coastlines were classified as continental.

2.6 Statistical analysis

Associations between environmental covariates and movement persistence were evaluated using a spatial generalized additive mixed model (GAMM). Prior to model fitting, all continuous covariates, including sea surface temperature, bathymetry, and distance to the nearest coast, were standardized by centering and scaling to unit variance. Coast type was included as a binary factor, with continental coast used as the reference level.

Let $\mu_i = E(\gamma_i)$ denote the expected movement persistence for observation i . We assumed

$$\gamma_i \sim \text{Beta}(\mu_i, \phi),$$

where $\mu_i \in (0, 1)$ denotes the mean movement persistence and ϕ is a precision parameter. The mean was linked to the predictors through a logit link, and the final model was specified as

$$\text{logit}(\mu_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 c_i + \beta_5 (x_{3i} \times c_i) + f(\text{lat}_i, \text{lon}_i) + b_{\text{id}(i)},$$

with

$$b_j \sim \mathcal{N}(0, \sigma_b^2),$$

where x_{1i} , x_{2i} , and x_{3i} denote standardized sea surface temperature, bathymetry, and distance to the nearest coast, respectively, and c_i denotes coast type. The term $f(\text{lat}_i, \text{lon}_i)$ represents a two-dimensional thin plate spline for spatial location with basis dimension $k = 50$, and $b_{\text{id}(i)}$ denotes the random intercept associated with the shark corresponding to observation i . This random effect was included to account for repeated observations within individuals. The model was fitted using the `mgcv` package in R with restricted maximum likelihood (REML). Model adequacy was assessed using residual diagnostics and basis-dimension checks.

A beta response with a logit link was used because movement persistence is bounded on the open interval $(0, 1)$. The shark-specific random effect accounts for within-individual dependence arising from repeated daily observations, whereas the two-dimensional spatial smooth captures broad residual geographic structure beyond that explained by the measured covariates. The resulting specification is consistent with the processed dataset, which consists of repeated daily movement estimates linked to environmental and coastal predictors. Interpretation remains associative rather than causal. The spatial smooth captures residual spatially structured variation in movement persistence after adjustment for the measured environmental and coastal covariates, and this remaining pattern may reflect unmeasured environmental, ecological, or geographic influences.

All analyses were performed in R version 4.5.1.

3 Results

3.1 Track processing and daily predictions

After initial data cleaning, the telemetry dataset contained 12,157 observations from 60 individual whale sharks. Track segmentation and filtering retained 58 segments from 32 sharks for subsequent state-space modeling. These retained segments were then regularized to daily predicted locations using the state-space framework described above.

Movement persistence was estimated from the resulting daily predicted trajectories, and environmental covariates were linked to each daily position. Trajectories for which movement persistence estimation did not converge were excluded from subsequent analysis. In addition, observations with positive extracted bathymetry values were removed because they corresponded to biologically implausible locations on land. After these processing steps, the final analysis dataset used in the spatial GAMM contained 1,509 daily observations. Of these, 1,140 were classified as nearest to continental coasts and 369 as nearest to island coasts. These retained observations formed the basis of the subsequent environmental analysis and represent the subset of tracks for which daily movement persistence and environmental covariates could be estimated reliably.

3.2 Summary of environmental and coastal covariates

The final analysis dataset included 1,509 daily predicted locations with linked environmental and coastal covariates. These covariates comprised sea surface temperature, bathymetry, distance to the nearest coast, and coast type. Coast type was unevenly represented in the final dataset, with 1,140 observations classified as nearest to continental coasts and 369 classified as nearest to island coasts.

Summary statistics for the continuous covariates are presented in Table 1. In the final analysis dataset, sea surface temperature had a mean of 25.42°C (SD 1.81) and ranged from 18.41 to 29.16°C. Bathymetry remained negative after filtering, with a mean of -3357.99 m (SD 975.51) and values ranging from -6276.65 to -0.49 m. Distance to the nearest coast had a mean of 667.04 km (SD 556.75), with values ranging from 0 to 2592.63 km. These summaries describe the environmental and coastal conditions represented in the modeled whale shark locations.

Table 1: Summary statistics for continuous environmental covariates in the final analysis dataset.

Variable	Mean	SD	Median	Min	Max
SST (°C)	25.42	1.81	25.68	18.41	29.16
Bathymetry (m)	-3357.99	975.51	-3574.57	-6276.65	-0.49
Distance to coast (km)	667.04	556.75	581.43	0.00	2592.63

3.3 Environmental and coastal effects on movement persistence

The final spatial GAMM identified several significant environmental and coastal correlates of whale shark movement persistence. On the standardized scale, movement persistence increased with higher sea surface temperature ($\beta = 0.140$, $p < 0.001$), shallower bathymetry ($\beta = 0.097$, $p = 0.007$), and greater distance to the nearest coast ($\beta = 1.645$, $p < 0.001$). These results indicate that more directional movement tended to occur in warmer waters, in relatively shallower marine environments, and farther from the nearest coastline. Coast type was not significant as a main effect ($\beta = 0.108$, $p = 0.344$), suggesting no overall difference in movement persistence between island- and continent-associated coasts after adjustment for the other predictors. The corresponding fitted associations on the response scale are shown in Figure 2.

A significant interaction was detected between standardized distance to coast and coast type ($\beta = -0.474$, $p < 0.001$). The positive association between distance to the nearest coast and movement persistence was therefore weaker for locations nearest to island coasts than for those nearest to continental coasts. Figure 3 shows the corresponding model-based predictions on the response scale. Predicted movement persistence increased with offshore distance in both settings, but the increase was steeper for continent-associated coasts. This indicates that coastal context modified the strength of the offshore association even though coast type alone was not significant as a main effect.

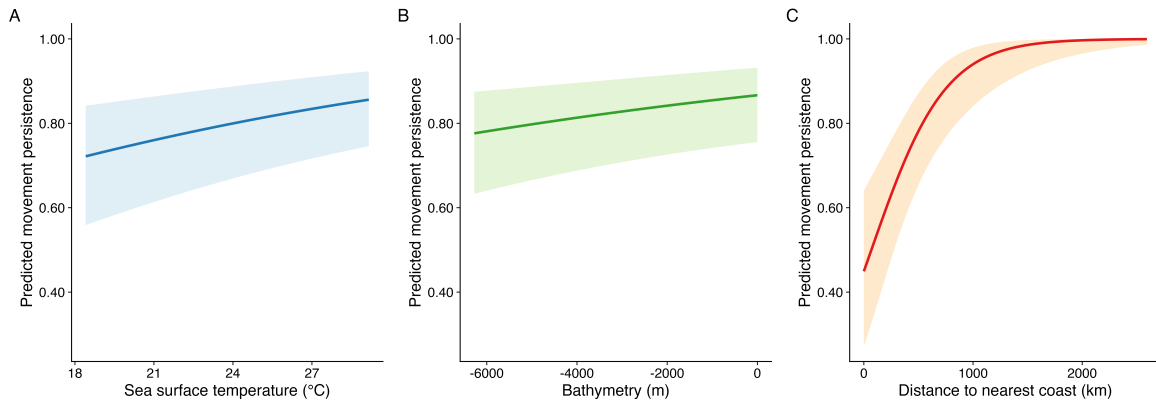


Figure 2: Marginal model-based associations between whale shark movement persistence and the three continuous predictors included in the final GAMM: sea surface temperature, bathymetry, and distance to the nearest coast. Solid lines represent fitted values and shaded bands represent 95% confidence intervals. Movement persistence increased with warmer sea surface temperature, shallower bathymetry, and greater offshore distance. Showing these associations on the response scale helps relate the standardized regression results to interpretable environmental gradients.

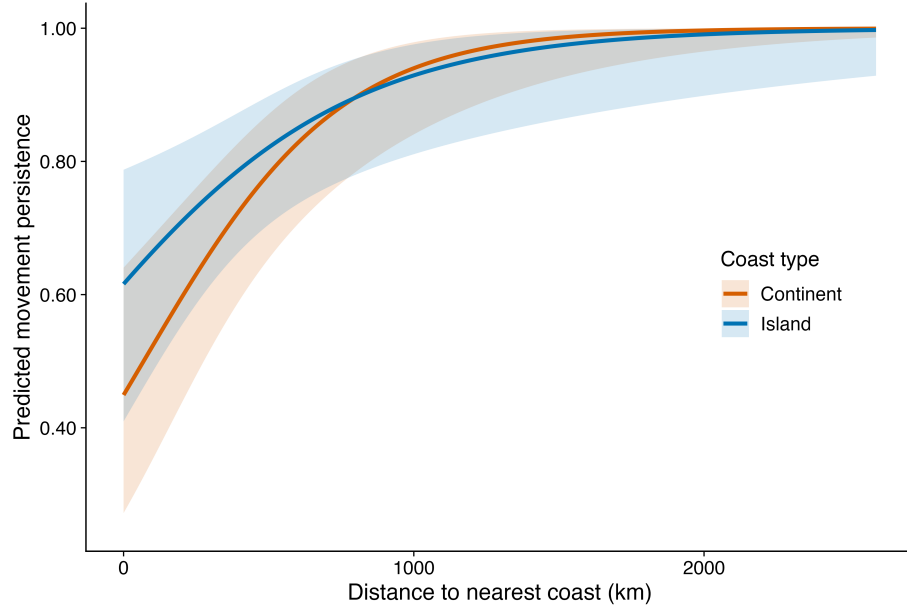


Figure 3: Model-based predicted movement persistence across distance to the nearest coast for island- and continent-associated coastal settings. Shaded bands represent 95% confidence intervals. Predicted movement persistence increased with offshore distance in both settings, but the increase was steeper for continent-associated coasts, consistent with the significant interaction between distance to coast and coast type. This indicates that coastal context modified the strength of the offshore association even though coast type alone was not significant as a main effect.

The shark-specific random effect was highly significant ($\text{edf} = 49.88$, $p < 2 \times 10^{-16}$), indicating substantial between-individual heterogeneity in baseline movement persistence. Overall, the model explained 85.3% of the deviance, with an adjusted R^2 of 0.793. A full summary of the parametric coefficients and smooth-term significance results from the final model is provided in the Supplementary Material.

3.4 Residual spatial structure in the spatial GAMM

In addition to the measured environmental and coastal predictors, the two-dimensional spatial smooth remained highly significant ($\text{edf} = 45.06$, $p < 2 \times 10^{-16}$), indicating marked residual geographic structure in movement persistence across the study region. Figure 4 shows the estimated spatial smooth from the final model after adjustment for sea surface temperature, bathymetry, distance to the nearest coast, coast type, and shark-level random effects.

Areas of relatively higher and lower fitted movement persistence were distributed unevenly across the eastern Pacific. This remaining structure suggests that broader spatial processes, or additional unmeasured environmental drivers, may still contribute to whale shark movement behaviour beyond the covariates included in the model. The strong significance and complexity of the spatial smooth indicate that accounting for geographic structure was an important component of the final modeling framework. Additional diagnostic plots for the final model are provided in the Supplementary Material and did not indicate major problems in the residual-fitted relationship, although some departure from normality was evident in the residual tails.

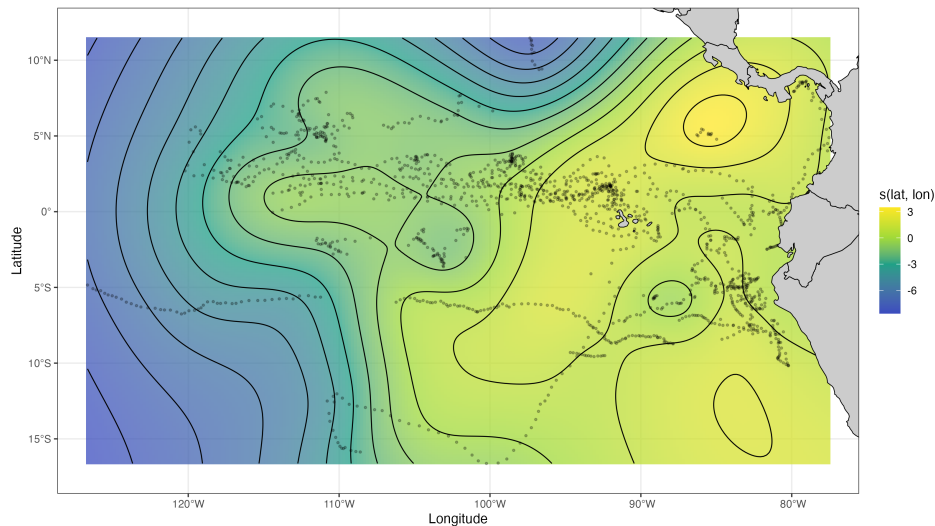


Figure 4: Estimated spatial smooth from the final GAMM across the eastern Pacific study region. Colours represent the fitted contribution of geographic location to whale shark movement persistence after adjustment for sea surface temperature, bathymetry, distance to the nearest coast, coast type and shark-level random effects. Warmer colours indicate areas with relatively higher fitted movement persistence, whereas cooler colours indicate lower fitted movement persistence. Contour lines highlight the complexity of the residual spatial structure, showing that substantial geographic variation remained after accounting for the measured covariates.

4 Discussion

This study identified several environmental and coastal correlates of whale shark movement persistence in the eastern Pacific. Movement persistence was higher in warmer waters, in shallower bathymetry, and at greater distances from the nearest coast. Although coast type was not significant as a main effect, the association between offshore distance and movement persistence differed between island- and continent-associated settings, indicating that coastal context modified this relationship. Residual spatial structure also remained after adjustment for the measured covariates, suggesting that important

geographic variation in movement persistence was not fully explained by the environmental and coastal predictors included here.

One possible interpretation of the positive association with sea surface temperature is that warmer waters may be associated with broader environmental conditions that favour more sustained directional movement. In this sense, sea surface temperature may reflect not only thermal conditions themselves, but also larger-scale habitat context within the study region. The positive association with shallower bathymetry may similarly indicate that movement persistence differed across underlying habitat structure, or that bathymetry acted partly as a proxy for other ecological or oceanographic features not directly measured in the present analysis.

The interaction between distance to coast and coast type further suggests that the offshore gradient did not have the same behavioural meaning across coastal settings. In particular, the steeper increase in movement persistence with offshore distance for continent-associated locations may indicate a stronger transition toward sustained directional movement away from continental coasts. The weaker offshore association near island coasts may instead reflect differences in coastline geometry, nearby habitat configuration, or local oceanographic conditions that were not explicitly represented in the model.

The remaining spatial structure after adjustment for the measured covariates also suggests that important sources of geographic variation were not fully captured by sea surface temperature, bathymetry, distance to coast, and coast type alone. This residual pattern may reflect additional environmental, ecological, or behavioural drivers, including spatial processes that were not directly observed in the available data. The spatial smooth should therefore be interpreted as capturing remaining geographic structure in movement persistence, rather than as representing a single directly identifiable mechanism.

Several limitations should be considered when interpreting these results. Regularization to a daily time scale was necessary for linking movement estimates to daily environmental covariates, but it may have smoothed over shorter-term movement variation present in the original telemetry records. In addition, track segmentation and filtering excluded long temporal gaps, short segments, non-converged trajectories, and locations with positive extracted bathymetry values, so the final analysis dataset represents a restricted subset of the original observations. Movement persistence should also be interpreted specifically as a continuous summary of directional consistency and does not provide a complete characterization of behavioural state. The environmental and coastal covariates were derived from external spatial datasets and nearest-feature calculations, which may introduce measurement error or simplification, particularly for coastline classification and extracted bathymetry. More broadly, the results are associative rather than causal. The remaining spatial structure should therefore be understood as residual geographic variation in movement persistence after adjustment for the measured covariates, and it may reflect additional environmental, ecological, or behavioural drivers not included in the model.

5 Conclusion

This analysis showed that whale shark movement persistence was associated with both environmental conditions and coastal context, while also exhibiting substantial residual geographic variation. Several next steps follow from these results. Future work could incorporate additional dynamic environmental covariates to assess whether the remaining spatial pattern is partly explained by unmeasured oceanographic conditions. It could also examine temporal heterogeneity more explicitly and evaluate whether individual-level variation in movement persistence extends beyond the random intercept structure used here. These extensions would help refine interpretation of the residual geographic variation identified in the present analysis.

A Supplementary Materials

A.1 Supplementary Methods

The final spatial GAMM was fitted in R using the `mgcv` package, with movement persistence modeled using a beta response and logit link. The model included standardized sea surface temperature, standardized bathymetry, standardized distance to the nearest coast, coast type, their interaction, a two-dimensional thin plate spline for spatial location, and a shark-level random intercept. Smoothing parameters were estimated by restricted maximum likelihood, and model adequacy was assessed using residual diagnostics and basis-dimension checks. The fitted model corresponded to the following implementation in R:

```
gam(
  gamma_beta ~ sst_z + bathymetry_z + dist_to_coast_km_z * coast_type +
    s(lat, lon, bs = "tp", k = 50) + s(id, bs = "re"),
  family = betar(link = "logit"),
  method = "REML"
)
```

A.2 Final spatial GAMM results

Table 2 reports the parametric coefficient estimates from the final spatial GAMM, and Table 3 reports the significance results for the smooth terms.

Table 2: Parametric coefficient estimates from the final spatial GAMM.

Term	Estimate	Std. Error	z value	p-value
Intercept	1.33972	0.22645	5.916	3.29×10^{-9}
sst_z	0.13978	0.03645	3.835	0.000125
bathymetry_z	0.09744	0.03608	2.700	0.006930
dist_to_coast_km_z	1.64468	0.39820	4.130	3.62×10^{-5}
coast_typeisland	0.10773	0.11385	0.946	0.344033
dist_to_coast_km_z $\times$ coast_typeisland	-0.47417	0.13023	-3.641	0.000272

Table 3: Approximate significance results for smooth terms in the final spatial GAMM.

Term	edf	Ref. df	Chi-square	p-value
$s(\text{lat}, \text{lon})$	45.06	48.02	1020	$< 2 \times 10^{-16}$
$s(\text{id})$	49.88	52.00	2491	$< 2 \times 10^{-16}$

A.3 Supplementary Figures

Additional diagnostic plots for the final spatial GAMM are shown in Figure 5.

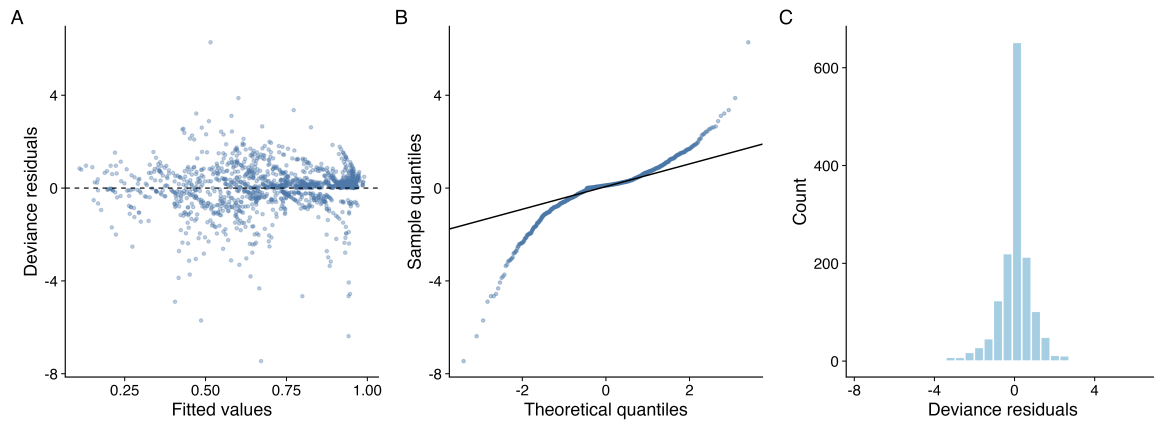


Figure 5: Diagnostic plots for the final spatial GAMM used to assess model adequacy. Panel A shows deviance residuals against fitted values, with residuals generally centered around zero and no strong systematic pattern across the fitted range. Panel B shows a normal Q–Q plot of deviance residuals, indicating noticeable departure from normality in the tails. Panel C shows the distribution of deviance residuals, which is concentrated near zero but includes some more extreme residual values. Together, these diagnostics suggest that the model captured the main structure in the data reasonably well, although some tail behaviour remained imperfectly fitted.

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